22/9/25

X : RV $\quad Y = g(X) \quad g: \mathbb{R} \to \mathbb{R}$

distbn of $X \Rightarrow$ distbn of Y

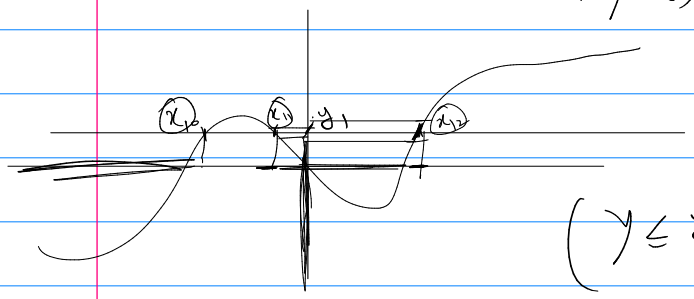
Discrete : "Tabulation" → Support of Y

$$p_Y(y) = \sum_{x:\, g(x)=y,\, x \in \text{Supp}(X)} p_X(x)$$

Continuous : → Support of Y

→ CDF method

$$F_Y(y) = Pr(Y \le y) = Pr\left(X \in \{x : g(x) \le y\}\right)$$



Example

$$(Y \le y_1) \longleftrightarrow (X \le x_{10}) \cup (x_{11} \le X \le x_{12})$$

$$f_Y(y) = F_Y'(y)$$

→ PDF method (g : smooth)

Solve $y = g(x)$ to get $x_1(y), x_2(y), \ldots$

Ex: $y = x^2 \quad x_1(y) = \sqrt{y}, \; x_2(y) = -\sqrt{y}$

Simple case: g : monotone

$$f_Y(y) = \sum_{\substack{i=1,2,\ldots \\ x_i(y) \in \text{Supp}(X)}} f_X(x_i(y)) \cdot \frac{1}{|g'(x_i(y))|}$$

$$= |(g^{-1})'(y)|$$

Continuous

Ex: (1) $Y = aX + b$

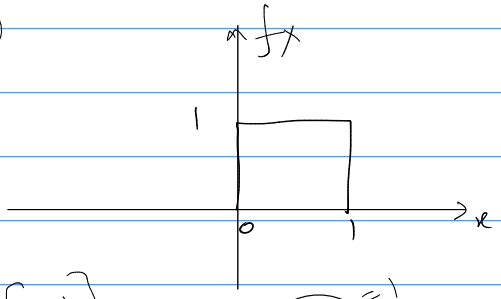
$$f_Y(y) = f_X\left(\frac{y-b}{a}\right) \cdot \frac{1}{|a|}$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \text{ all } x$$ "Gaussian" (mean 0, var 1)

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\,|a|} e^{-\frac{(y-b)^2}{2a^2}}, \text{ all } y$$ "Gaussian" + mean b, var a^2

(2) $Y = X^2$

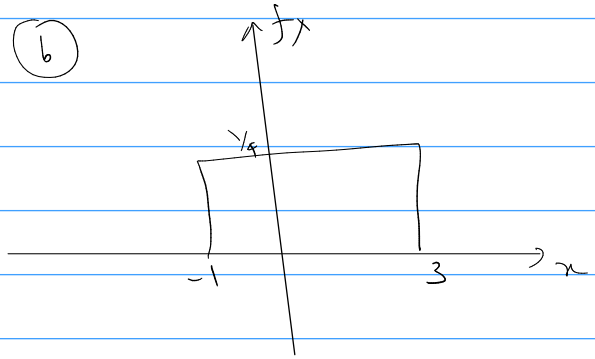
(a)



$y \in [0,1]$

$$f_Y(y) = f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}}, \quad y \in (0,1)$$

(b)



$y \in [0,9]$

$y \in [0,1]$ $\sqrt{y}$ & $-\sqrt{y}$

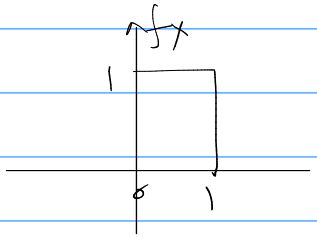
$$f_Y(y) = \frac{1}{4} \cdot \frac{1}{\sqrt{y}}, \quad y \in [0,1]$$

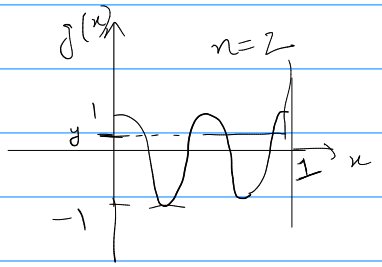
$y \in [1,9]$ $\sqrt{y}$

$$f_Y(y) = \frac{1}{4} \cdot \frac{1}{2\sqrt{y}}, \quad y \in [1,9]$$

(3) $Y = \cos 2\pi n X$

$n = 1, 2, 3, \ldots$





$$f_Y(y) = 1 \cdot \frac{1}{\pi \cdot \sqrt{1-y^2}} \cdot 2^n, \quad -1 < y < 1$$

---

2 var $\longrightarrow$ 2 var $\qquad X, Y \sim f_{XY}$

PDF $\qquad (x,y) \longrightarrow (u,v)$

$u(x,y)$, $v(x,y)$ } smooth $\qquad$ inverse $x(u,v)$, $y(u,v)$

$$J = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$du\,dv = |\det(J)|\,dx\,dy$$

$$\tilde{J} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

One inverse

$$f_{UV}(u,v) = f_{XY}(x(u,v), y(u,v)) \cdot \frac{1}{|\det(J)|}$$

evaluated at $x(u,v), y(u,v)$

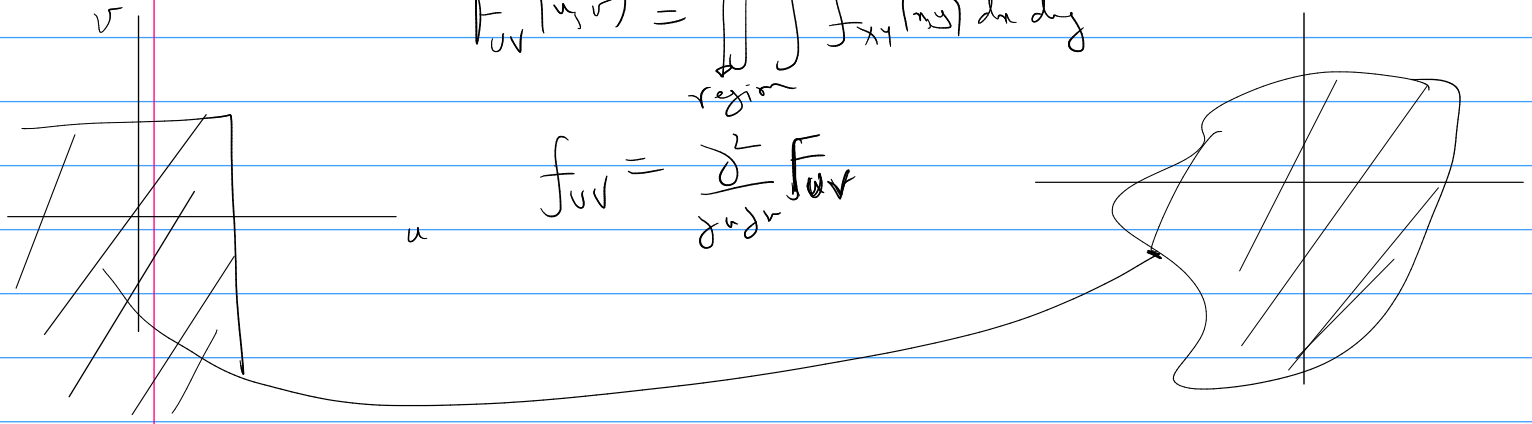
$$= |\det(\tilde{J})|$$

CDF

$(U \le u,\ V \le v) \longleftrightarrow$ Suitable region in (x,y)

$$F_{UV}(u,v) = \iint_{\text{region}} f_{XY}(x,y)\,dx\,dy$$

$$f_{UV} = \frac{\partial^2}{\partial u \partial v} F_{UV}$$



Ex: ①

$$U = aX + bY \qquad X = \frac{1}{(ad-bc)}(dU - bV)$$

$$V = cX + dY \qquad Y = \frac{1}{(ad-bc)}(-cU + aV)$$

$$J = |ad-bc|$$

$ad \neq bc$

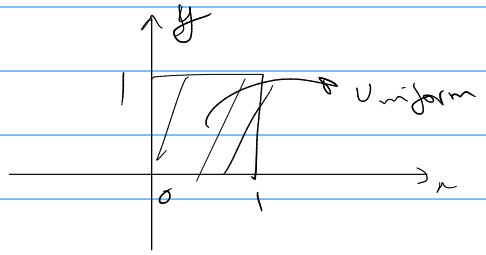
②

$$f_{XY}(x,y) = \frac{1}{2\pi} e^{-x^2/2} e^{-y^2/2} \quad \text{all } x,y$$

$$= \frac{1}{2\pi} e^{-\frac{(du-bv)^2 + (-cu+av)^2}{2(ad-bc)^2}}$$

$$f_{UV}(u,v) = \frac{1}{2\pi|ad-bc|} \cdot e^{-\frac{(du-bv)^2 + (-cu+av)^2}{2(ad-bc)^2}}, \text{ all } u,v$$

b



$$U = X + Y \qquad X = \frac{U+V}{2}$$

$$V = X - Y \qquad Y = \frac{U-V}{2}$$

$$|\det(J)| = 2$$

$$f_{XY} = \begin{cases} 1, & 0 < x,y < 1 \\ 0, & \text{else} \end{cases}$$

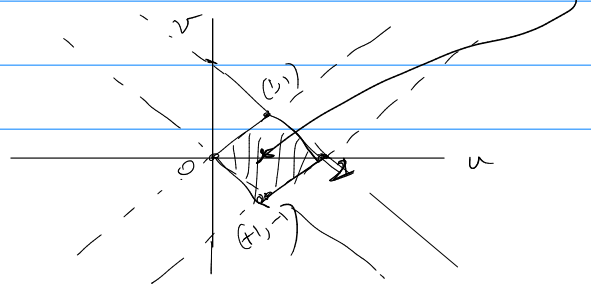
$$0 < x < 1$$

$$0 < y < 1$$

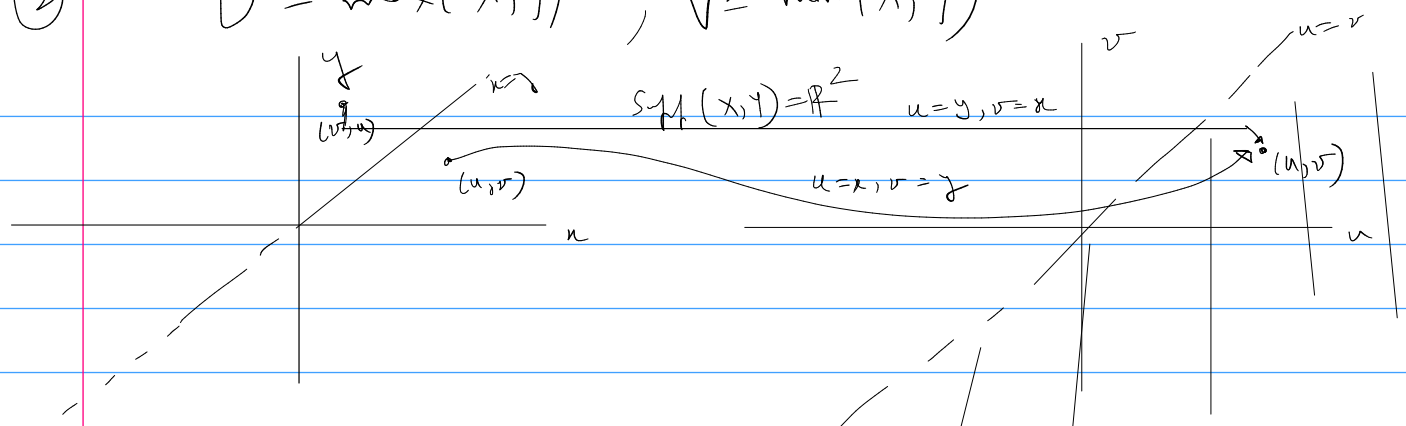
$$f_{UV}(u,v) = \frac{1}{2}, \ (u,v)$$

$$0 < \frac{u+v}{2} < 1 \qquad u+v > 0,\ u+v < 2$$

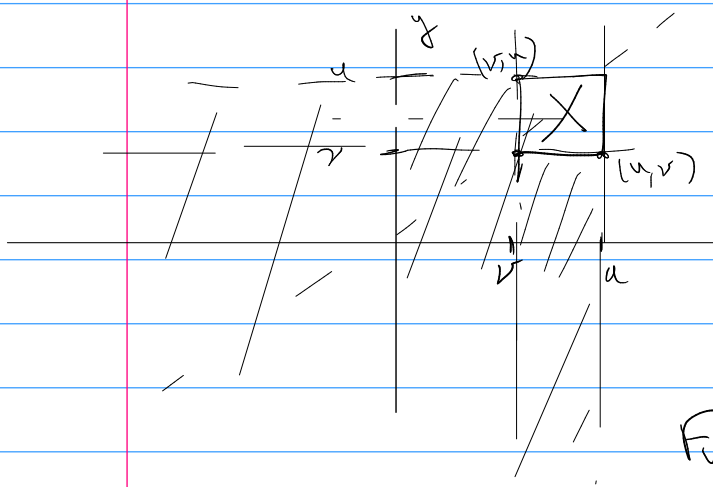
$$0 < \frac{u-v}{2} < 1 \qquad u-v > 0,\ u-v < 2$$



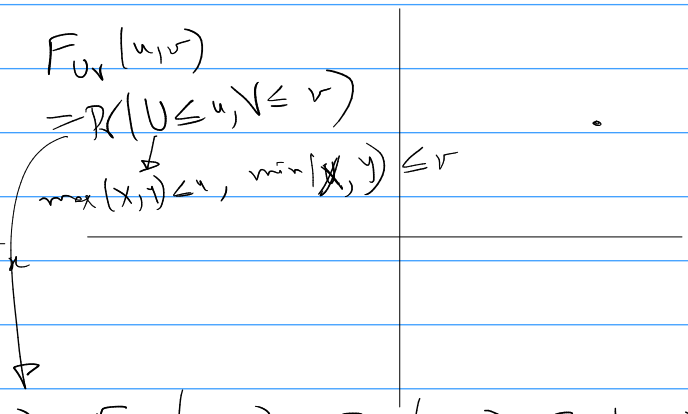
(2) $U = \max(X,Y)$, $V = \min(X,Y)$



$$f_{UV}(u,v) = f_{XY}(u,v) + f_{XY}(v,u), \quad u > v$$



$F_{UV}(u,v)$
$= P(U \le u, V \le v)$
$\max(X,Y) \le u, \min(X,Y) \le v$



$$F_{UV}(u,v) = F_{XY}(u,v) + F_{XY}(v,u) - F_{XY}(v,v)$$